

Homework Assignment #5

Professor: Eric Gerber

Name: _____

Instructions: Please include the following information on the first page of your completed homework write-up:

1. Your name
2. DS 4420
3. Homework #5

You will submit **up to four files** to Gradescope for this homework:

- Handwritten/latex typeset answers may be included in their own **.pdf** file or as part of either of the below
- A **.ipynb** file with all python code/output
- A **.pdf** file knitted from an included **.rmd** with all R code/output

Answers that are not supported by reasoning/work will not receive full credit. **Homework is due by 11:59 pm, via Gradescope, on the date above.** Late submissions will **not** be accepted, but you may receive extra credit for early submission (see syllabus for details).

You will also be graded on organization/neatness of the submitted files.

Theory Warm Up (40 points)

(1: 10 points)

Given:

$$y|\theta, n \sim \text{Bin}(n, \theta) \Rightarrow p(y|\theta, n) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}, \quad y \in \{0, 1, \dots, n\}$$

$$\theta \sim \text{Beta}(\alpha, \beta) \Rightarrow p(\theta) = \frac{\theta^{\alpha-1} (1 - \theta)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 \leq \theta \leq 1$$

Show **by hand** that the Beta is a conjugate prior for θ of the Binomial distribution.

(2: 10 points) Recall the probability function (pdf) for a Beta random variable, x :

$$p(x|\alpha, \beta) = \frac{x^{\alpha-1} (1 - x)^{\beta-1}}{B(\alpha, \beta)}$$

We have previously defined the Beta function, $B(\alpha, \beta) = \int_0^1 t^{\alpha-1} (1 - t)^{\beta-1} dt$, but we can also solve this function using the **Gamma** function, $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$:

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

It turns out the Gamma function, $\Gamma(z)$ is very simple if z is an integer: $\Gamma(z) = (z - 1)!$.

Given $X \sim \text{Beta}(\alpha = 2, \beta = 2)$, **prove (by hand, integrating the pdf)** that $E(X) = .5$.

(3: 20 points) Complete the following problems **by hand**.

(a) Given:

$$y|\theta \sim \text{Expon}(\theta) \Rightarrow p(y|\theta) = \theta e^{-\theta y}, \quad 0 \leq y < \infty$$

$$\theta \sim \text{Gamma}(\alpha, \beta) \Rightarrow p(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta}, \quad 0 \leq \theta < \infty$$

Show that the Gamma is a conjugate prior for θ of the Exponential distribution. To help you (and to speed things along), note that in this case:

$$\int_0^\infty \theta^\alpha e^{-\theta(y+\beta)} d\theta = \frac{\Gamma(\alpha+1)}{(y+\beta)^{\alpha+1}}$$

(b) Given a collection of samples from an exponential distribution, $y_i|\theta \sim \text{Expon}(\theta)$, find the posterior mean, $E(\theta|y_i)$ given a Gamma prior distribution with hyperparameters $\alpha = \beta = 1$.

(c) Given:

$$y|\theta \sim \text{Poiss}(\theta) \Rightarrow \frac{\theta^y e^{-\theta}}{y!}, \quad y \in \{0, 1, \dots\}$$

$$\theta \sim \text{Gamma}(\alpha, \beta) \Rightarrow p(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta}, \quad 0 \leq \theta < \infty$$

Show that the Gamma is a conjugate prior for θ of the Poisson distribution. To help you (and to speed things along), note that in this case:

$$\int_0^\infty \theta^{y+\alpha-1} e^{-\theta(1+\beta)} d\theta = \frac{\Gamma(y+\alpha)}{(1+\beta)^{y+\alpha}}$$

(d) Given a collection of samples from a Poisson distribution, $y_i|\theta \sim \text{Poiss}(\theta)$, find the posterior mean, $E(\theta|y_i)$ given a Gamma prior distribution with hyperparameters $\alpha = \beta = 1$.

Single Parameter Models (60 points)

For the problems in this section, you will make use of the **ds4420_spotify.csv** file on Canvas.

(4: 30 points) You are interested in finding the posterior distribution for θ , the probability a song is explicit or not. In **R**, write your own functions for the prior and posterior distributions of the probability (θ) a song is explicit. Assume each song follows a Bernoulli distribution, and with a Beta prior with hyperparameters $\alpha = 2$ and $\beta = 18$. Then:

- i. Plot the prior and posterior distributions on top of each other.
- ii. Based on the hyperparameters, what was our prior belief about the value of θ ? Add a red vertical line to the plot from (i) corresponding to this value.
- iii. Calculate the *MLE* for θ . Add a blue vertical line to the plot from (i) corresponding to this value.
- iv. Calculate the posterior mean of θ , $E(\theta|y)$. Add a purple vertical line to the plot from (i) corresponding to this value.
- v. What do you note about the relationship between these three measures? Discuss **in a couple sentences** how you would generate predictions for whether a song is explicit or not, and how you would validate the performance of this model.

THE HOMEWORK CONTINUES ON THE NEXT PAGE

(Extra Credit: 5 points) Complete problem 4 in **Python**.

(5: 30 points) You are interested in finding the posterior distribution for μ , the average popularity of a song, assuming that a track's popularity is $y_i | \mu, \sigma \sim N(\mu, \sigma = 20)$. The conjugate prior for a Gaussian mean μ is also Gaussian (the proof of this is annoying) such that:

$$\mu \sim N(\eta, \tau)$$

And the posterior distribution is:

$$\mu | y \sim N(\mu^*, \sigma^*)$$

Where:

$$\mu^* = \frac{\frac{n\bar{y}}{\sigma^2} + \frac{\eta}{\tau^2}}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}} \quad \text{and} \quad \sigma^* = \sqrt{\left(\frac{n}{\sigma^2} + \frac{1}{\tau^2}\right)^{-1}}$$

(a) To proceed, set $\sigma = 20$, $\eta = 50$, $\tau = 2$. In **R**, write your own functions for the prior and posterior distributions of the mean (μ) popularity of a song. Then complete the steps on the next page:

- i. Plot the prior and posterior distributions on top of each other.
 - To help with reading the visualization, I recommend the x-axis range of 40 to 80.
- ii. Based on the hyperparameters, what was our prior belief about the value of μ ? Add a red vertical line to the plot from (i) corresponding to this value.
- iii. Calculate the *MLE* for μ . Add a blue vertical line to the plot from (i) corresponding to this value.
- iv. What is the posterior mean of μ , $E(\mu | y)$? Add a purple vertical line to the plot from (i) corresponding to this value.
- v. What do you note about the relationship between these three measures?

(b) Repeat part (a) using two more prior distributions (make sure to keep all output for both):

1. $\mu \sim N(\eta = 75, \tau = 2)$
2. $\mu \sim N(\eta = 50, \tau = 5)$

(c) Discuss **in detail**:

- i. Which prior distribution resulted in the more significantly different posterior from the model in (a)?
- ii. What (if anything) does this exercise tell you about the how the hyperparameters of the prior distribution are used in Bayesian modeling?
- iii. How would you generate predictions for how popular a song is, and how you would validate the performance of these models.

(Extra Credit: 5 points) Complete problem 5 in **Python**.